

## Review: Integer operations



1 Evaluate the following:

a  $9 + (-7) - 4$

c  $20 + (-14) - (-2) - 18$

e  $-18 + 21 \div 7$

g  $(3 - 15) \times (-14 + 9)$

i  $(-36) \div (-4) - 14$

k  $-7 + 8 \times 6 \div 2$

m  $(-100) \div (-4) \div (-5)$

o  $(120 \div 5) \times 3$

q  $9 \times (-6) + 28 \div (-7)$

s  $2 \times (-25 - 20) \div (-7 + 2)$

b  $-9 + 6 - (-17)$

d  $6 - (13 - 3)$

f  $-4 \times (-7 + 18)$

h  $70 - 8 \times (-7)$

j  $(-390) \div (-19 + 6)$

l  $54 \div 3 \div (-2)$

n  $72 \div 6 \div 2$

p  $6 - 25 \div 5 + 4$

r  $(-14) \div 2 - 10 \div (-2)$

2 Complete the following equation with  $<$ ,  $>$  or  $=$  to make the statement true.

$$6 + (3 + (-11)) \text{ } \mathbb{I} - 2 + (4 + (-6))$$

3 State the first step to simplify the following expressions:

a  $-6 \div 2 \times 4$

b  $-14 \div (5 - 3 \times 4)$

4 Complete the following statements to make each of the following equations true:

a  $-3 \times 4 - \mathbb{I} = -14$

b  $2 - (-3 \times \mathbb{I}) = 17$

c  $\mathbb{I} + 5 \times 2 = 4$

d  $(-17 - 3) \div \mathbb{I} = -4$

---

## Additional questions

5 Valentina dives off a platform that is 8 m above the water. She descends 14 m before returning to the surface.

a If Valentina's initial position on the diving platform is represented by the integer 8, find the integer that represents her lowest position throughout the dive.

b Hence, state the greatest depth that Valentina reaches during the dive.

c One evening in Moscow the temperature fell to  $-3^{\circ}\text{C}$ . By midday the next day the



- 7 Liquid nitrogen can be used to make ice cream. The ice cream mixture starts at  $73^{\circ}\text{F}$ , then rapidly cools to  $-301^{\circ}\text{F}$  after liquid nitrogen is added to it. State the integer that represents this change in temperature.
- 8 Three contestants are heading into the final round of a game show. Eileen has 39 points and Roxanne has  $-3$  points. If Oprah has 14 more points than Eileen, calculate how many points separate Roxanne and Oprah.
- 9 A submarine is 148 m below sea level. It rises 34 m, then dives 47 m. Find the distance of the submarine below sea level.
- 10 Each bottle of orange juice is supposed to contain 250 mL. Nine bottles are inspected by the quality assurance department. The results of the inspection are shown in the table:

| Bottle                                         | 1  | 2  | 3   | 4  | 5  | 6 | 7  | 8  | 9  | 10 |
|------------------------------------------------|----|----|-----|----|----|---|----|----|----|----|
| Amount below or above the required volume (mL) | +4 | -7 | -11 | +9 | +3 | 0 | -1 | -5 | +6 | +1 |

- a Find the actual volume of:
- i Bottle 1      ii Bottle 2      iii Bottle 6      iv Bottle 10
- b Calculate the total volume contained in the 10 bottles.
- 11 Each letter has been assigned to an integer. A word must be formed using these letters and each letter can only be used once. The table shows the assignment of the integers:

| Letter  | A  | E  | I  | C | T | N | R | S |
|---------|----|----|----|---|---|---|---|---|
| Integer | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 |

The score of the word is the sum of the integers assigned to its letters. Find the score of the following words:

- a NET      b EAR      c RAIN      d CARES
- 12 Are the following statements true or false? Explain.
- a The product of a positive number with a negative number is negative.
- b The product of an even number of negative numbers is negative.



**13** Is the quotient of a positive integer and a negative integer positive or negative? Explain.

**14** Find the value of  $n$  in the following statements:

- a** A number  $n$  is multiplied with  $-3$  to give 21.
- b** A number  $n$  is multiplied with  $-10$  to give  $-40$ .

**15** Find the value of  $n$  in the following statements:

- a** A number  $n$ , when divided by 3, gives  $-9$ .
- b** A number  $n$  when divided by  $-5$  gives  $-4$ .

**16** Which quadrant(s) could the point  $(x, y)$  lie in if:

- a**  $xy > 0$
- b**  $\frac{x^2}{y} < 0$

**17** In golf your score for each hole is determined by comparing it to the par (or expected) number of shots for that hole. Taking one less shot than expected will give a score of  $-1$ , one more shot will give  $+1$ , and so on.

Ivan played 18 holes of golf, and the frequency of each score he obtained is shown in the adjacent table:

| Score | Frequency |
|-------|-----------|
| $-1$  | 5         |
| Par   | 3         |
| $+1$  | 4         |
| $+2$  | 6         |

- a** Find Ivan's total score for his 6 scores of  $+2$ .
- b** Find Ivan's total score for his 5 scores of  $-1$ .
- c** Find Ivan's total score for the total 18 holes.

**18** Victoria earns \$1100 a month, and deposits it into her bank account. She also withdraws \$200 a month to pay for rent.

- a** How much, in total, does Victoria deposit into her bank account in one full year?
- b** How much, in total, does Victoria withdraw from her bank account in one full year?
- c** Find the total change in her bank account over the year. An increase will be represented by a positive number and a decrease will be represented by a negative number.

**19** Explain why the following statement is incorrect, and find the correct answer:



20

- a The temperature in Paris is  $13^{\circ}\text{C}$  degrees Celsius less than double the temperature in London. If it is  $8^{\circ}\text{C}$  in London then what is the temperature in Paris?
- b The temperature in Sydney is  $21^{\circ}\text{F}$  more than triple the temperature in New York. If it is  $9^{\circ}\text{F}$  in New York then what is the temperature in Sydney?

21 Laura and a group of 11 friends are ordering some pizza. They think that they will eat two pizzas for every three people. Pizzas cost \$14 each and there is a \$8 delivery fee. If they split the total cost evenly between each person, how much will Laura need to pay?

22 At the start of the week, Quiana had \$131 in her bank account. On Monday, she withdrew \$43. On Wednesday, she deposited \$11, and on Thursday, she deposited three times as much as she deposited on Wednesday. Finally, she withdrew \$25 on Friday. Find Quiana's final bank balance at the end of the week.

23 Starting from his house, Jerry walked  $1\text{ m/s}$  due south for 30 minutes, then  $2\text{ m/s}$  due north for 15 minutes and then, then due south again  $2\text{ m/s}$  for 45 minutes. Find Jerry's final distance from his house.